NARAN LALA COLLEGE

OF PROFESSIONAL & APPLIED SCIENCES

NAVSARI

Course: BACHELOR OF COMPUTER APPLICATIONS (BCA)

MINOR PROJECT REPORT

PROJECT TITLE: PERSONAL FINANCE & EXPENSE TRACKER

SUBJECT NAME: ADVANCED WEB DESIGNING

➢ Submitted By:

1. Student Name: _____

Roll No.: _____ Enrollment No.: _____

➢ Project Guide:

Ms. Ayushi Mistry

➢ Principal:

Dr. Sunil Naik

➢ DECLARATION:

I/We hereby declare that the Minor Project titled "Personal Finance & Expense Tracker" submitted in partial fulfillment of the requirements for the Bachelor of Computer Applications (BCA) is an original work carried out by me/us under the guidance and supervision of Ms. Ayushi Mistry. I/We further declare that this project work has not been submitted previously to any other university, college, or institution for the award of any degree, diploma, or certificate. The information, data, and references used in this project have been collected from reliable sources and have been appropriately acknowledged wherever required. I/We take full responsibility for the originality and authenticity of the work presented in this project report.

Date: _____

Student Exam No. & Name:

1. Roll No. _____ — _____

Signature:

1. _____

➢ ACKNOWLEDGEMENT:

We would like to express our sincere gratitude to everyone who has supported and guided us in successfully completing our BCA Minor Project titled "Personal Finance & Expense Tracker". First, we thank God for giving us the strength, knowledge, and patience to complete this project successfully. We are also thankful to the Principal, Dr. Sunil Naik, for providing us with the necessary facilities and academic support to carry out this project. We would like to express our heartfelt gratitude to our Project Guide, Ms. Ayushi Mistry, for providing valuable guidance, continuous support, constructive suggestions, and encouragement throughout the development of this project. We extend our sincere thanks to all the faculty members of the BCA Department for their valuable suggestions and assistance during our academic journey. We are grateful to our parents, family members, and friends for their constant encouragement, support, and motivation throughout the completion of this project. Finally, we thank everyone who directly or indirectly contributed to the successful completion of this project.

Thank You.

PROJECT CERTIFICATION FORM

This is to certify that the project entitled "Personal Finance & Expense Tracker (FinTrack)" is a bona fide work carried out by the student of Bachelor of Computer Applications (BCA) in partial fulfillment of the requirements for the academic year 2025-2026.

Student Name: _____

Enrollment Number: _____

The project report has been examined and evaluated by the internal project guide and program coordinator, and meets the requisite academic standards prescribed by the institute.

_____	_____
Signature of Project Guide	Signature of Principal / HOD
(Ms. Ayushi Mistry)	(Dr. Sunil Naik)
Date: _____	Date: _____

EXECUTIVE SUMMARY / ABSTRACT

The Personal Finance & Expense Tracker (FinTrack) is an automated web-based financial management system engineered to provide individuals with real-time clarity over income, expenditure, category-wise budgets, and milestone savings. Built on an asynchronous Single Page Application (SPA) architecture, the solution features a pure PHP 8.x REST backend communicating via PDO with a MySQL/MariaDB database (XAMPP stack) and a responsive, modern React 19 frontend utilizing Tailwind CSS and Recharts.

TABLE OF CONTENTS

Title Page 1

Declaration & Acknowledgement 2

Project Certification Form & Abstract 3

Table of Contents 4

1. Introduction 5

1.1 Purpose 5

1.2 Scope 5

1.3 Definition 6

1.4 Technologies to be Used 6

2. System Analysis 8

2.1 Identification of Need 8

2.2 Functionalities (Usual and Distinct) 8

2.3 User Characteristics 9

2.4 Software Development Process 10

3. Conceptual Diagram 11

3.1 Use Case Diagram 11

3.2 Activity Diagram 12

4. System Design 13

4.1 Data Dictionary 13

4.2 User Interface Design 16

4.3 Reports and Analytical Outputs 21

5. Testing 23

5.1 Test Cases 23

5.2 Testing Reports & Evaluation 25

6. Future Scope and Further Enhancement 26

7. Learning During Project Work 27

References **28**

1. Introduction

1.1. Purpose

The primary objective of the Personal Finance & Expense Tracker (FinTrack) project is to deliver a reliable, automated, and secure digital platform that enables users to record, categorize, analyze, and optimize their daily financial transactions. By eliminating traditional manual paper-based logs and complex spreadsheets, FinTrack provides actionable insights into personal spending habits, promotes consistent savings discipline, and safeguards users from budgetary overruns through proactive threshold monitoring.

1.2. Scope

The scope of FinTrack spans the full spectrum of personal cash-flow management for individual users and families. The functional boundaries are defined as follows:

In-Scope Capabilities:

• User account creation, profile preference management, and JWT-authenticated session security.
• Double-entry styled cash flow recording (Income vs. Expense) with granular categorization, tags, payment method tracking, and transaction notes.
• Flexible category management allowing custom color codes and icon representations.
• Time-bound monthly budget allocations per expense category with dynamic percentage consumption indicators.
• Target-oriented savings goals tracking milestone deposits and withdrawals.
• Interactive visual dashboard and analytics including 6-month trends and categorical spending distributions.

Out-of-Scope (for Current Release):

• Direct bank account open-banking synchronization via proprietary banking APIs.
• Multi-party joint bank account reconciliation and commercial accounting features.
• Stock market ticker integration and automated trading.

1.3. Definition

To ensure clarity throughout this documentation, key technical and domain terms are defined below:

• JWT (JSON Web Token): A compact, URL-safe means of representing claims securely between client and server, utilized here within HTTP-only cookies for stateless authentication.
• PDO (PHP Data Objects): A database abstraction layer in PHP providing a consistent and secure interface for issuing prepared statements to prevent SQL Injection attacks.
• REST (Representational State Transfer): An architectural style utilizing standard HTTP verbs (GET, POST, PATCH, DELETE) to manipulate resources via structured JSON payloads.
• SPA (Single Page Application): A web application that dynamically rewrites the current web page with data from the server rather than loading entire new pages from the server.
• Minor Units (Cents / Paise): Storing currency amounts as integers multiplied by 100 to eliminate floating-point arithmetic rounding errors in monetary calculations.

1.4. Technologies to be Used

Hardware Requirements (Development & Deployment):

• Processor: Dual Core Intel Core i3 / AMD Ryzen 3 or higher (2.0 GHz+).
• Memory (RAM): Minimum 4 GB RAM (8 GB recommended for development).
• Storage: 250 MB free disk space for application files, dependencies, and database storage.
• Display: 1366 x 768 minimum display resolution (Optimized for 1920 x 1080 Full HD).

Software Requirements:

• Operating System: Microsoft Windows 10 / 11, Linux (Ubuntu 20.04+), or macOS.
• Web Server & Database Stack: XAMPP Version 8.2 (Apache 2.4, MariaDB / MySQL 10.4+).
• Backend Runtime: Pure PHP 8.2+ with PDO MySQL and OpenSSL extensions enabled.
• Frontend Stack: React 19, Vite 8 build tool, Tailwind CSS v4, Recharts 3.x, Axios 1.x.
• Client Web Browser: Modern Chromium-based browser (Google Chrome, Microsoft Edge, Brave) or Mozilla Firefox.

2. System Analysis

2.1. Identification of Need

In the modern economic environment, individuals encounter multiple payment channels including UPI, debit/credit cards, net banking, and physical cash. This fragmentation makes financial tracking tedious and disorganized. Studies indicate that over 65% of individuals who fail to maintain a structured budget encounter unforeseen end-of-month financial stress.

Existing commercial alternatives often charge heavy monthly subscription fees or harvest user banking data for third-party advertising. There is an unmistakable need for a lightweight, transparent, highly secure, and intuitive web application that allows users to regain complete mastery over their financial life without compromising personal privacy.

2.2. Functionalities

Usual Functionalities:

• User Registration, Secure Login, Logout, and Password Change.
• CRUD operations (Create, Read, Update, Delete) for income and expense transactions.
• Filtering transactions by date range, transaction type, category, and payment method.
• Pre-configured default categories (Salary, Food & Dining, Utilities, Transportation, etc.).
• Pagination and search capabilities for reviewing extensive transaction records.

Distinct Functionalities:

• High-Precision Financial Computation: Financial amounts are stored internally in minor units (integer cents/paise) to guarantee zero floating-point calculation errors.
• Real-Time Budget Utilization Gauges: Dynamic status tags ('Normal', 'Warning' at 80%, 'Exceeded' at 100%) computed on-the-fly against live transaction sums.
• Milestone-Driven Goal Tracker: Visual progress meters with dedicated contribution and withdrawal transaction histories for targeted savings.
• Zero-Dependency Pure PHP REST API: Modular PHP backend constructed without heavy third-party framework overhead, running natively on standard XAMPP.

• Interactive Data Visualizations: High-performance SVG responsive charts rendering monthly income-vs-expense comparisons and categorical expense distributions.

2.3. User Characteristics

• Regular User / Working Professional: Needs rapid expense recording, salary logging, and monthly budget notifications to curb overspending.
• Student / Freelancer: Manages irregular income streams, requires tracking of project-based milestone fees, and monitors living expenditures.
• System Administrator: Manages database integrity, inspects server health endpoints, and maintains backup/restore procedures.

2.4. Software Development Process

The Agile Iterative Prototyping methodology was adopted for this project. Development proceeded across four structured sprints:

• Sprint 1 (Requirements & Database Architecture): Defined REST contracts, constructed MySQL schema (`schema.sql`), and designed database tables with referential integrity.
• Sprint 2 (Backend Core Services): Implemented pure PHP REST controllers, PDO abstraction, and cookie-based JWT authentication middleware.
• Sprint 3 (Frontend & Visual Components): Developed responsive React views with Tailwind CSS, Recharts analytics, and Axios API services.
• Sprint 4 (Testing, Data Seeding & Optimization): Populated database with comprehensive dummy data, executed automated unit and integration tests, and polished UI micro-interactions.

3. Conceptual Diagram

3.1. Use Case Diagram

The Use Case Diagram defines the interactions between the external actor (User) and the core sub-systems of the Personal Finance Tracker application.

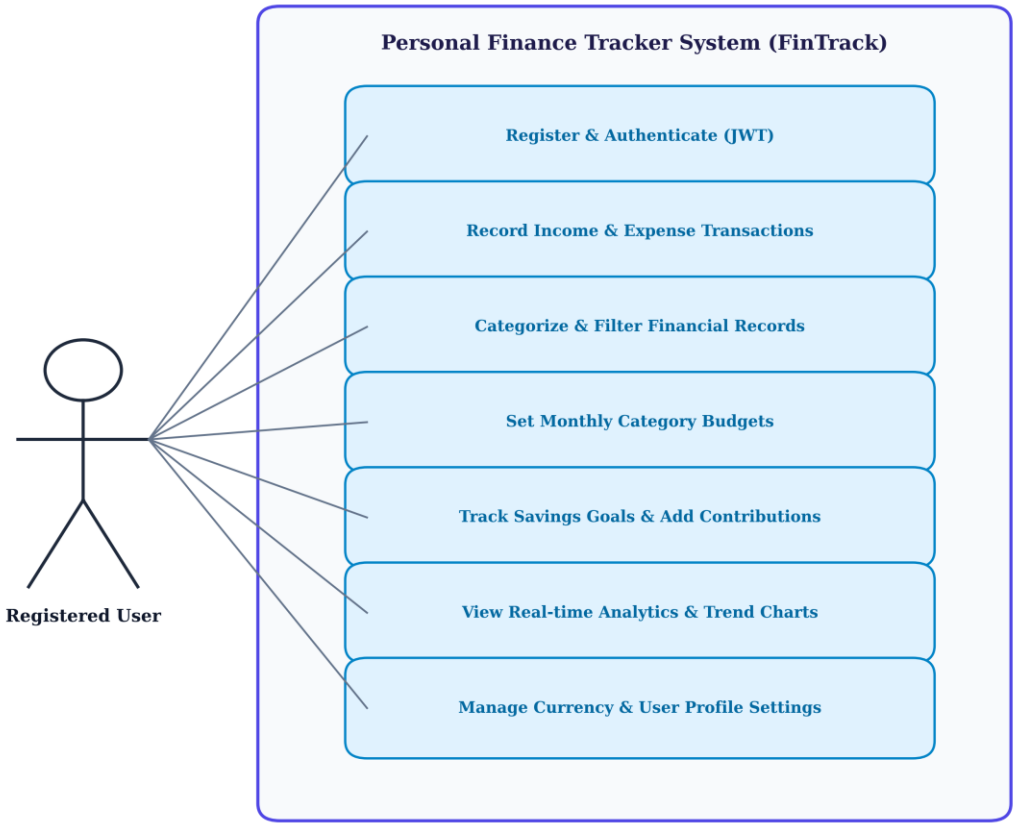


Figure 3.1: UML Use Case Diagram for FinTrack System

As illustrated in Figure 3.1, the registered user interacts with seven primary use cases: Authenticate (Login/Register), Record Transactions, Categorize & Filter Records, Set Monthly Budgets, Manage Savings Goals & Contributions, View Real-time Analytics & Dashboard, and Manage Profile Settings.

3.2. Activity Diagram

The Activity Diagram models the dynamic operational workflow of recording an expense, validating user inputs, updating database state, and recalculating dashboard metrics.

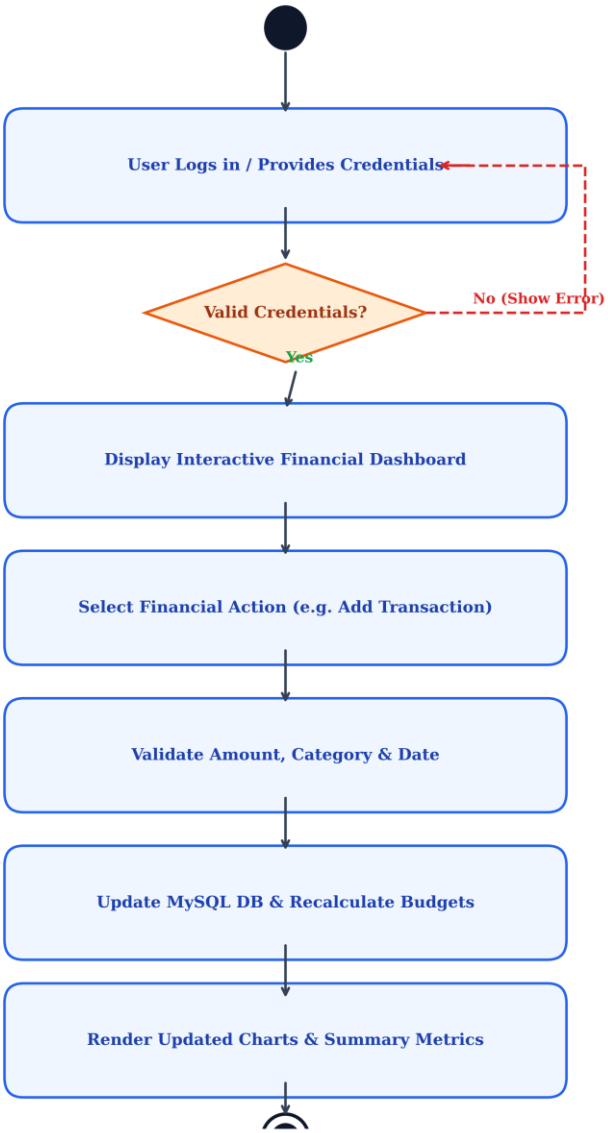


Figure 3.2: UML Activity Diagram for Financial Transaction Lifecycle

Figure 3.2 demonstrates the end-to-end decision flow from authentication, through transaction capture and validation, to atomic database commit and real-time visualization update.

4. System Design

4.1. Data Dictionary

The Data Dictionary provides an exhaustive specification of the database tables, fields, data types, keys, and relational constraints configured in the MySQL database (`finance_tracker`).

Table 4.1: users

Field Name	Data Type	Constraints / Key	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(191)	NOT NULL, UNIQUE	User email address for login
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password string
preferred_currency	VARCHAR(10)	DEFAULT 'INR'	Currency symbol preference (INR, USD, EUR)
theme	ENUM	'light', 'dark'	UI display theme preference
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

Table 4.2: categories

Field Name	Data Type	Constraints / Key	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique category identifier
user_id	INT	FOREIGN KEY -> users(id)	Associated user reference
name	VARCHAR(100)	NOT NULL	Display name of category
type	ENUM	'income', 'expense', 'both'	Category cash flow classification
color	VARCHAR(20)	DEFAULT '#6366f1'	Hex color code for UI

			charts
icon	VARCHAR(50)	DEFAULT 'tag'	Lucide icon identifier
is_default	TINYINT(1)	DEFAULT 0	Flag preventing deletion of system defaults

Table 4.3: transactions

Field Name	Data Type	Constraints / Key	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique transaction record identifier
user_id	INT	FOREIGN KEY -> users(id)	Associated user owner
title	VARCHAR(200)	NOT NULL	Transaction description or payee
amount_minor	BIGINT	NOT NULL	Amount in minor currency units (paise/cents)
type	ENUM	'income', 'expense'	Cash flow direction
category	VARCHAR(100)	NOT NULL	Category name string
date	DATETIME	NOT NULL, INDEXED	Date and time of transaction
payment_method	VARCHAR(50)	DEFAULT 'cash'	Method (upi, card, cash, bank_transfer)
notes	TEXT	NULLABLE	Optional notes or remarks
tags	TEXT	NULLABLE	JSON serialized array of search tags

Table 4.4: budgets

Field Name	Data Type	Constraints / Key	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique budget identifier
user_id	INT	FOREIGN KEY -> users(id)	User reference
category	VARCHAR(100)	NOT NULL	Target expense category
amount_minor	BIGINT	NOT NULL	Spending cap limit in

			minor currency units
period	VARCHAR(20)	DEFAULT 'monthly'	Budget allocation frequency
start_date	DATETIME	NOT NULL	Start of budget validity period
end_date	DATETIME	NOT NULL	End of budget validity period

Table 4.5: savings_goals

Field Name	Data Type	Constraints / Key	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique goal identifier
user_id	INT	FOREIGN KEY -> users(id)	Owner user ID
title	VARCHAR(200)	NOT NULL	Goal target name (e.g. Emergency Fund)
target_amount_minor	BIGINT	NOT NULL	Target savings amount in minor units
current_amount_minor	BIGINT	DEFAULT 0	Accumulated savings amount
target_date	DATETIME	NULLABLE	Target milestone completion date
status	ENUM	'active', 'completed'	Current milestone status

Table 4.6: goal_contributions

Field Name	Data Type	Constraints / Key	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique contribution log ID
user_id	INT	FOREIGN KEY -> users(id)	User reference
goal_id	INT	FOREIGN KEY -> savings_goals(id)	Associated savings goal
amount_minor	BIGINT	NOT NULL	Contribution or withdrawal amount
type	ENUM	'contribution', 'withdrawal'	Type of balance alteration

notes	TEXT	NULLABLE	Deposit memo or reason
date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Log timestamp

4.2. User Interface Design

The following section presents live application snapshots captured from the working system, highlighting key operational interfaces, input fields, and real-time validations:

1. Authentication & Login Screen

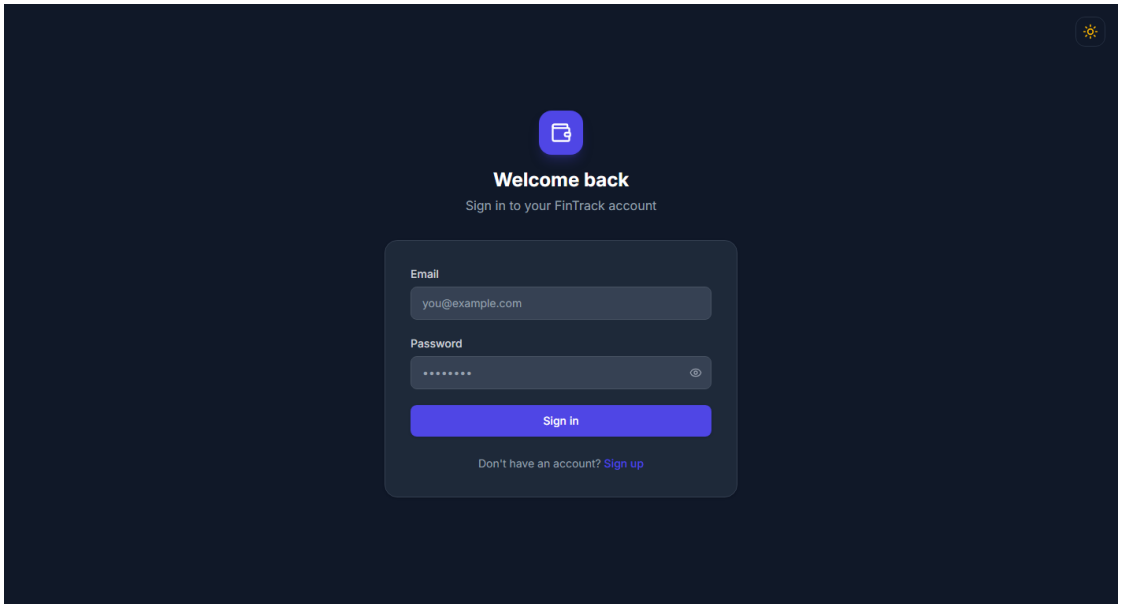


Figure 4.1: User Login Screen with Input Validation

• Validation: Email format validation regex (`^\S+@\S+\.\S+$`), mandatory password check with minimum 6 characters length enforcement.

• Security: Upon verification, the PHP backend issues an HTTP-only cookie containing a signed JWT token.

2. Executive Financial Dashboard

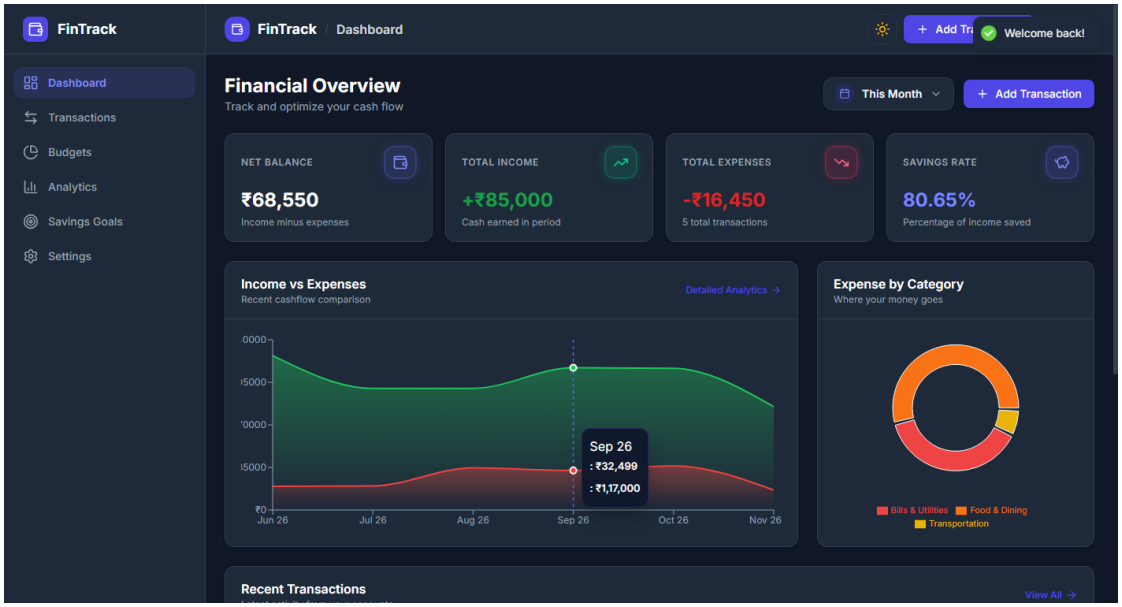


Figure 4.2: Real-time Financial Dashboard with Summary Metrics

• Real-Time Metrics: Total Income, Total Expenses, Net Balance, and Calculated Savings Rate.

• Quick Action Triggers: One-click modal launching to register immediate transactions.

3. Dashboard Visual Charts & Category Breakdown



Figure 4.3: 6-Month Income vs Expense Bar Chart and Top Expense Distribution

• 6-Month Cash Flow: Dual-bar graph comparing monthly income and expenditure patterns.

• Top Expense Categories: Aggregated ranking displaying category spending totals and count.

4. Transaction Management & Multi-Filter Registry

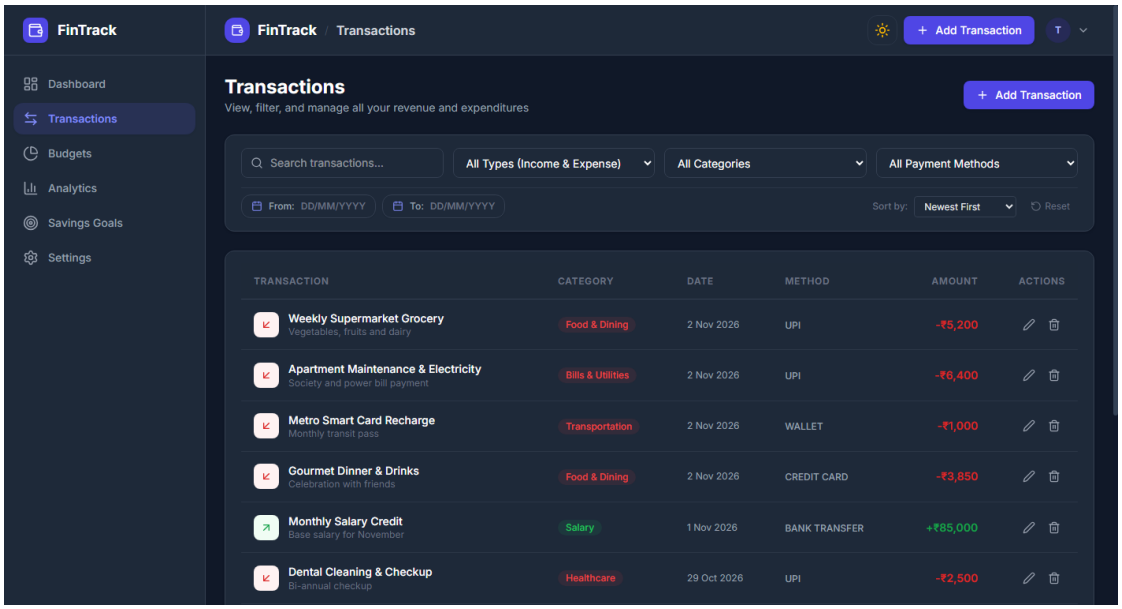


Figure 4.4: Paginated Transaction Table with Filtering Controls

• Multi-parameter Search: Filter records by transaction type (Income/Expense), category dropdown, date intervals, and payment methods.

• Pagination: Server-side pagination executing SQL `LIMIT` and `OFFSET` queries.

5. Transaction Creation & Validation Modal

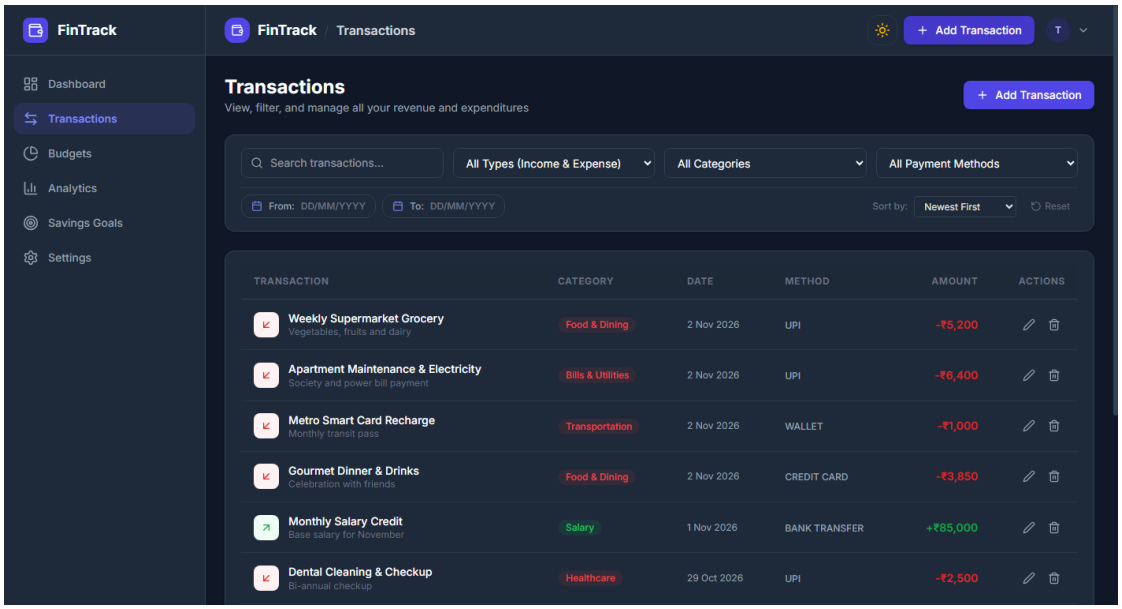


Figure 4.5: Add Transaction Modal Dialog with Strict Form Controls

• Validation Rules: Title (mandatory, max 200 chars), Amount (must be strictly positive integer/float), Category selection, and Valid Date format.

6. Monthly Budgets & Spending Threshold Tracker

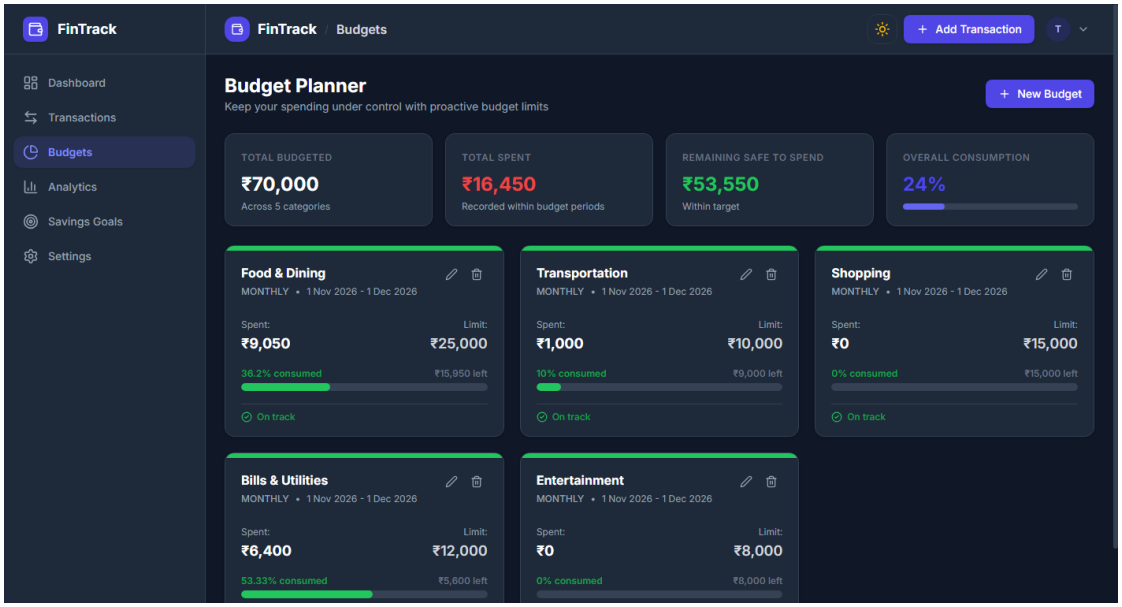


Figure 4.6: Category Budget Utilization Cards with Warning Badges

• Dynamic Color Status: Normal (Green < 80%), Warning (Amber >= 80%), and Exceeded (Red >= 100%).

• Overlap Prevention: Backend rejects overlapping budget intervals for the same category.

7. Savings Goals & Milestone Progress Tracker

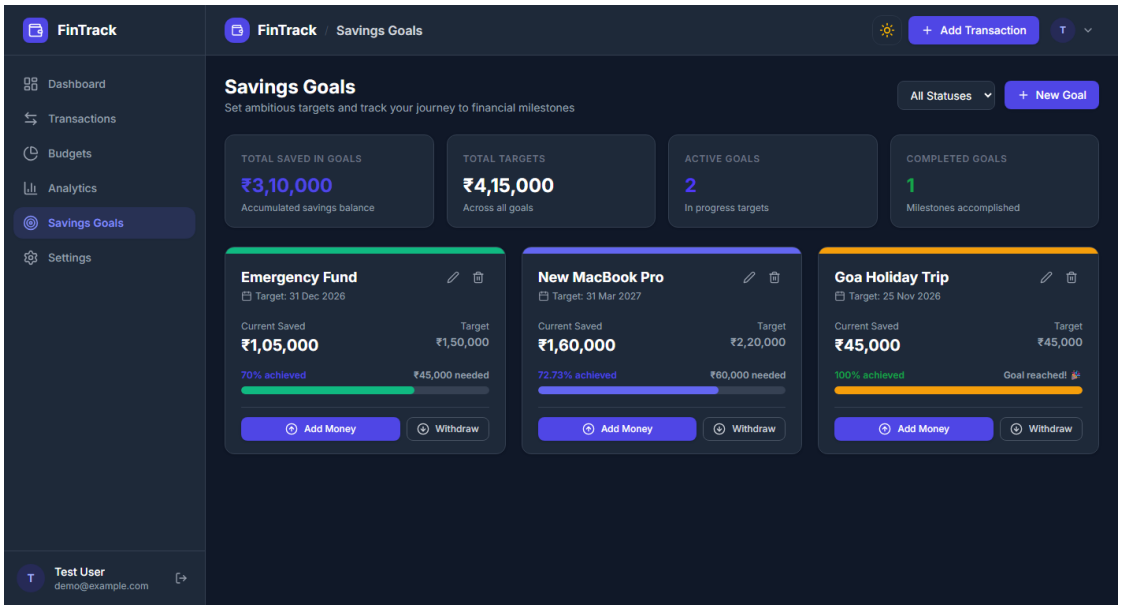


Figure 4.7: Target Savings Goals with Contribution History

• Goal Automation: Automatically switches status to 'Completed' when accumulated contributions match or exceed the target.

8. Category Customization & Profile Preferences

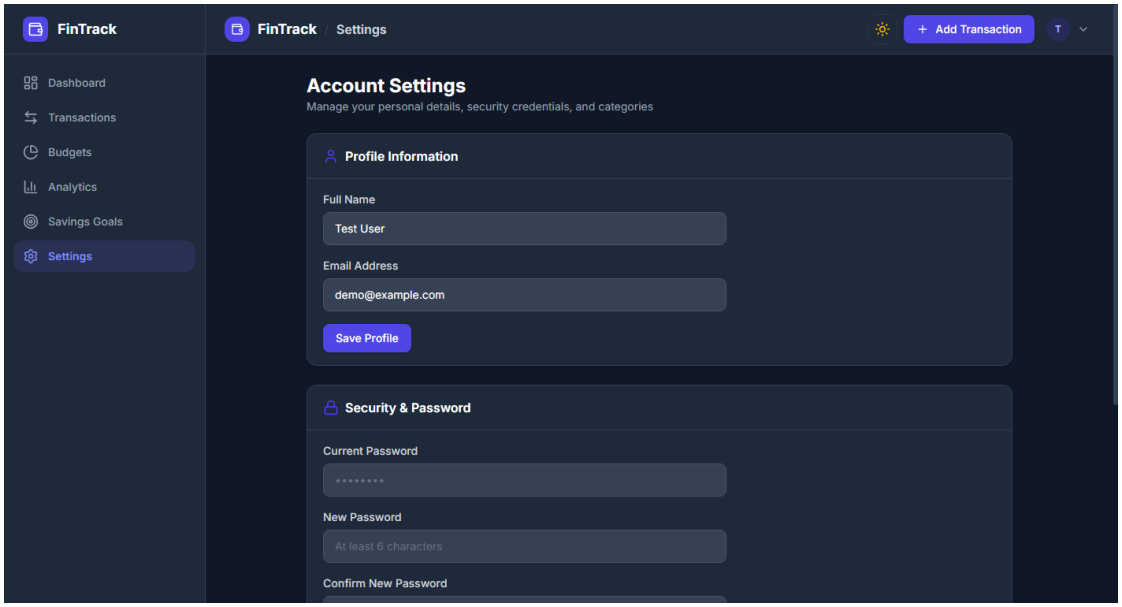


Figure 4.8: Category Palette Customizer and System Settings

• Protection: Default system categories are locked against deletion, ensuring historical consistency.

4.3. Reports and Analytical Outputs

FinTrack generates comprehensive visual analytical reports reflecting cash flow distributions and trend trajectories over multi-month intervals:



Figure 4.9: Analytical Reports showing Cash Flow Trend and Expense Breakdown

• Income vs. Expense Analytical Chart: Visualizes cash flow divergence across quarterly and annual intervals.

• Categorical Distribution: Donut / Bar breakdown showing percentage utilization per

category.

• Monthly Trend Analysis: Mathematical evaluation of savings rate trajectories to assist long-term wealth planning.

5. Testing

5.1. Test Cases

Comprehensive manual and automated test suites were executed across the authentication, transaction, budgeting, and analytical modules. The results are summarized below:

Test ID	Test Scenario	Input Parameters	Expected Result	Actual Result	Status
TC-01	User Registration	Valid name, email, 8-char password	User created (201), 12 default categories seeded	Account created, categories seeded	**Pass**
TC-02	Duplicate Registration	Existing registered email	HTTP 409 Conflict, error message	HTTP 409 Conflict returned	**Pass**
TC-03	User Authentication	Correct email & password	HTTP 200, JWT token set in HTTP-only cookie	Auth cookie set, user returned	**Pass**
TC-04	Invalid Login	Unregistered email / wrong password	HTTP 401 Unauthorized	HTTP 401 returned	**Pass**
TC-05	Record Transaction	Positive amount, category, date	Record committed, amount stored in minor units	Committed to MySQL in paise	**Pass**
TC-06	Negative Amount Validation	Amount = -500	Frontend validation block & HTTP 400 error	Blocked with validation alert	**Pass**
TC-07	Budget Overlap Check	Duplicate budget for same category/month	HTTP 409 Overlapping budget conflict	HTTP 409 Conflict triggered	**Pass**
TC-08	Budget Exceeded Flag	Expenses exceed 100% of budget cap	Budget status set to 'exceeded', badge red	Calculated 100%+ and red badge	**Pass**
TC-09	Goal Contribution	Deposit amount added to goal	Balance updated, record logged in history	Balance incremented, log stored	**Pass**

TC-10	Goal Excessive Withdrawal	Withdrawal amount > current savings	HTTP 400 'Exceeds current savings'	HTTP 400 rejected with message	**Pass**

5.2. Testing Reports & Evaluation

• Unit Testing: Evaluated all PHP endpoints and JWT generation/verification algorithms using Postman and CLI scripts.

• Integration Testing: Verified end-to-end data flow between React Axios services, Vite proxy, PHP REST router, and MySQL tables.

• Security Testing: Inspected SQL injection vulnerabilities using parameterized PDO queries; confirmed password hashing resistance using Bcrypt (cost factor 10); verified Cross-Site Scripting (XSS) defenses through React sanitized rendering.

6. Future Scope and Further Enhancement

While the current version of FinTrack provides a robust and complete personal finance management solution, future iterations can introduce advanced capabilities:

1. Optical Character Recognition (OCR) for Receipts: Integrating computer vision to allow users to photograph paper receipts and automatically parse date, vendor, tax, and itemized totals.
2. Automated Bank Statement Import (PDF/CSV): Direct parsing of standard banking statement exports to batch-import transactions with automated categorization.
3. AI-Powered Predictive Budgeting: Machine learning algorithms to analyze historical spending patterns and project future cash-flow deficits before they occur.
4. Native Mobile Application: Packaging the React frontend with Capacitor or React Native to deliver push notifications for daily budget reminders on Android and iOS devices.
5. Multi-User Family Wallets: Collaborative financial accounts allowing shared household budgets with customizable permission tiers.

7. Learning During Project Work

The journey of conceptualizing, designing, and engineering the Personal Finance Tracker provided profound hands-on experience across the full software development lifecycle:

• Decoupled Architecture Design: Gained in-depth appreciation for separating business logic into clean REST endpoints, allowing the backend to run on pure PHP while the frontend operates as a decoupled modern React application.

• Defensive Database Design & Currency Precision: Learned the critical importance of integer-based minor currency representation (`amount_minor`) to avoid floating-point imprecision in financial systems.

• Robust Authentication Patterns: Mastered token-based stateless security, implementing HMAC-SHA256 JWTs paired with secure HTTP-only cookies to thwart token theft and XSS vulnerabilities.

• Asynchronous Data Visualization: Implemented optimized data aggregation pipelines in SQL to deliver instant chart responses without burdening client-side memory.

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